Example 8.2

Calculate the value of Q_1, Q_3, D_6 and P_{50} from the following data

Roll No	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

Solution:

Marks are arranged in ascending order

12 15 20 28 30 40 50

n = number of observations =7

$$Q_1 = \text{Size of } \left(\frac{n+1}{4}\right)^{th} \text{ value}$$

$$= \text{Size of } \left(\frac{7+1}{4}\right)^{th} \text{ value}$$

$$= \text{Size of } 2^{nd} \text{ value } = 15$$

$$Q_3 = \text{Size of } \left(\frac{3(n+1)}{4}\right)^{th} \text{ value}$$

$$= \text{Size of} \left(\frac{3\times 8}{4}\right)^{th} \text{ value}$$

$$= \text{Size of } 6^{th} \text{ value} = 40$$

$$D_6 = \text{Size of } \left(\frac{6(n+1)}{10}\right)^{th} \text{ value}$$

$$= \text{Size of } \left(\frac{6\times 8}{10}\right)^{th} \text{ value}$$

$$= \text{Size of } 4.8^{th} \text{ value}$$

$$= \text{Size of } 5^{th} \text{ value} = 30$$

$$P_{50} = \text{Size of } \left(\frac{50(n+1)}{100}\right)^{th} \text{ value}$$

$$= \text{Size of } 4^{th} \text{ value} = 28$$

Hence Q_1=15, Q_3=40, D_6 =30 and P_{50}=28

(ii) Grouped data (discrete case):

Steps:

1. Arrange the data in ascending or descending order of magnitude.
2. Find out cumulative frequencies.
3. Apply the formula:

$$Q_1 = \text{Size of } \left(\frac{N+1}{4}\right)^{th} \text{ value}$$

$$Q_3 = \text{Size of } \left(\frac{3(N+1)}{4}\right)^{th} \text{ value}$$

Now look at the cumulative frequency (*cf*) column and find that total which is either equal to $\left(\frac{N+1}{4}\right)$ or next higher than that and determine the value of the variable corresponding to this. That gives the value of Q_1. Similarly Q_3 is determined with reference of $\frac{3(N+1)}{4}$ value of the variable.

Example 8.3

Compute Q_1, D_2 and P_{90} from the following data

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

Solution:

Marks X	Frequency f	Cumulative Frequency cf
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	N= 43

Table : 8.1

$$Q_1 = \text{Size of } \left(\frac{N+1}{4}\right)^{th} \text{ value}$$

$$= \text{Size of } \frac{43+1}{4} = 11^{th} \text{ value} = 20.$$

$$D_2 = \text{Size of } \left(\frac{2(N+1)}{10}\right)^{th} \text{ value}$$

$$= \text{Size of } \frac{88}{10} = 8.8^{th} \text{ value} = 20.$$

$$P_{90} = \text{Size of } \left(\frac{90(N+1)}{100}\right)^{th} \text{ value}$$

$$= \text{Size of } \frac{3960}{100} = 39.6^{th} \text{ value} = 50.$$

(iii) Grouped data (Continuous case):

In the case of continuous frequency distribution, the classes are arranged either in ascending or descending order and the

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